

MA415 Group8 Final Project Report
Analysis of the impact of COVID 19 on MBTA ridership in Boston
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Introduction

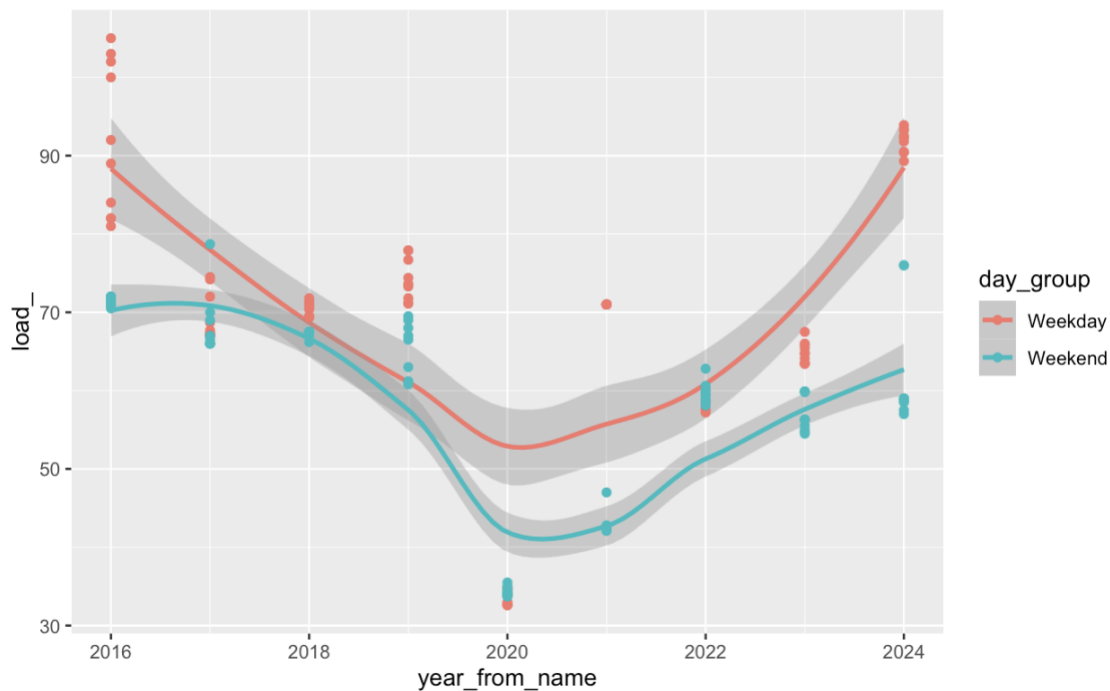
The COVID-19 had a huge impact on people's daily lives in Boston during 2020. More than 15,000 people died and over half a million residents contracted the disease in Massachusetts alone. In response, the Massachusetts state government declared a state of emergency, closed schools and many non-essential businesses, and issued a stay-at-home advisory[1]. As a result, people significantly changed their daily routines: most individuals stayed home, worked remotely, and took online classes. These changes greatly affected the local transportation system, as fewer people were willing or required to use public transit during that time. In this report, we will use the Boston MBTA bus ridership dataset from 2016 to 2024 to analyze the impact of COVID-19 on Boston's public transportation system. We employ several methods, such as line graphs, boxplots, and comparative visualizations to investigate which bus routes were most popular before the pandemic and how ridership patterns changed during and after COVID-19.

In our data analysis, we first reclassified the entire dataset into four time periods: "Pre-COVID" (all data before 2020), "COVID-2020" (all data during the year 2020, marked as the "COVID period"), "Recovery Period" (all data between 2021–2022), and "Post-COVID" (data after 2023). We also separated weekday and weekend ridership into two different categories to allow for clearer comparisons of bus route popularity across different time periods and COVID phases.

Analysis 1: Top 10 Loads Each Year

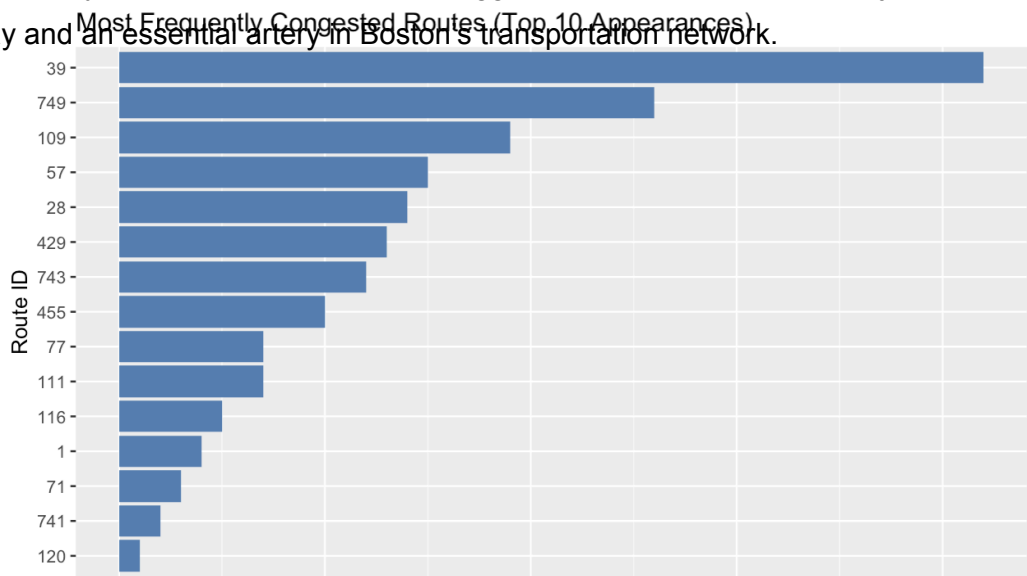
The first graph shows the top 10 bus loads in Boston for each year from 2016 to 2024. Overall, both weekday and weekend ridership follow a similar pattern: a gradual decline leading up to 2020, a sharp drop in 2020 during the COVID-19 pandemic, and a steady recovery beginning in 2021. This pattern reflects the significant reduction in travel caused by the pandemic, when stay-at-home advisories, remote work, and school closures dramatically limited people's need to use public transportation. The extremely low ridership around 2020 for both weekdays and weekends also suggests a decline in tourism and non-essential travel during the height of the pandemic. After 2020, weekday ridership appears to recover more quickly than weekend ridership. This difference implies that work-related travel began to resume earlier than leisure or

social travel, indicating a gradual transition from remote work back to in-person work for many riders.



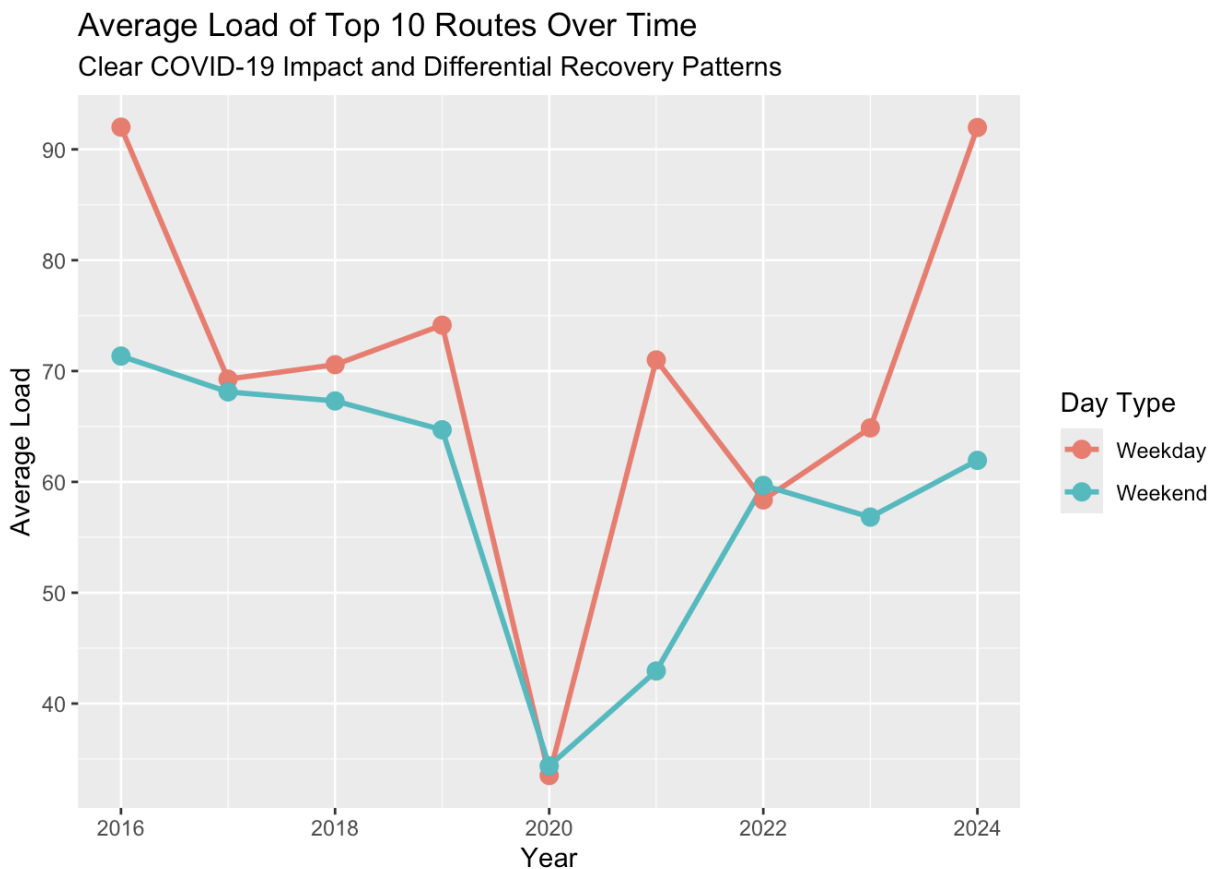
Analysis 2: Most Frequent Routes in Top 10

The second graph shows the bus routes that appear most frequently in the yearly Top 10 highest-load routes. From the graph, Route 39 stands out as the most frequently congested route, appearing almost twice as often as Route 749, which ranks second. According to the MBTA route information, Route 39 runs from Forest Hills, one of Boston's major transit hubs and a connection point for the Orange Line, toward central Boston(Fenway area)[2]. This helps explain its consistently high ridership: Forest Hills serves large residential neighborhoods, and many commuters use Route 39 to travel inbound to work, school, and commercial areas. The high frequency of Route 39 in the Top 10 suggests that this corridor is a major commuter pathway and an essential artery in Boston's transportation network.



Analysis 3: Average Load Trends

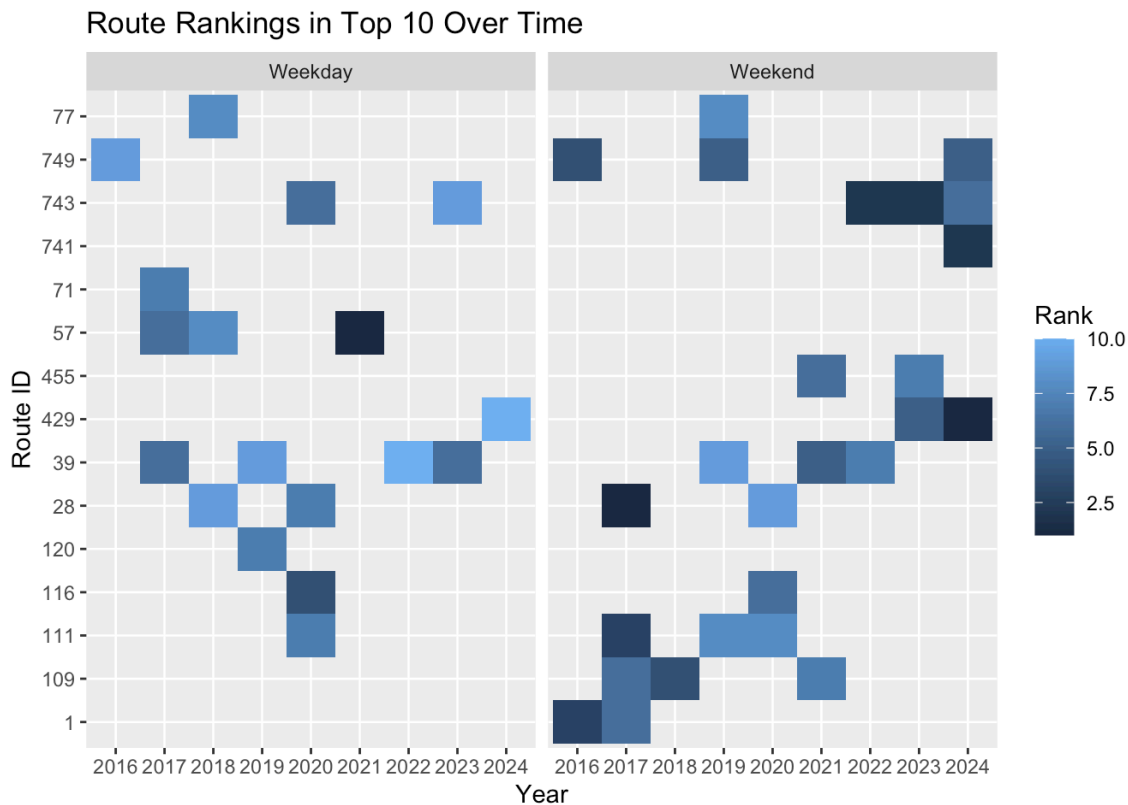
The third graph tracks the average load of the top 10 busiest routes from 2016 to 2024, which is separated by weekdays and weekends. From 2016 to 2019, both weekday and weekend averages are relatively stable, which shows that the demand on the core high-ridership routes was strong. However, in 2020, the average load for both day types drops sharply which is due to the collapse in bus usage during lockdowns, and work from home that reduced travel. After 2020, the lines begin to rise again, but not at the same speed. Weekend averages recover more quickly than weekday ones, which shows that discretionary trips came back sooner than those regular commuting patterns. Even by 2024, the average load on top routes has not fully returned to its pre-pandemic levels, which implies that some changes in travel behavior may be persistent rather than temporary, or the MBTA needs more time to recover.



Analysis 4: Route Rankings Over Time (Heatmap)

This heatmap shows how the rankings of the top 10 busiest MBTA routes change from 2016 to 2024. A few routes like 1, 28, 39, and 57 appear frequently, which shows that they are consistently high-demanded. Around 2020, rankings became more scattered, which is because COVID-19 disrupted normal travel patterns. Weekends show more fluctuation than weekdays

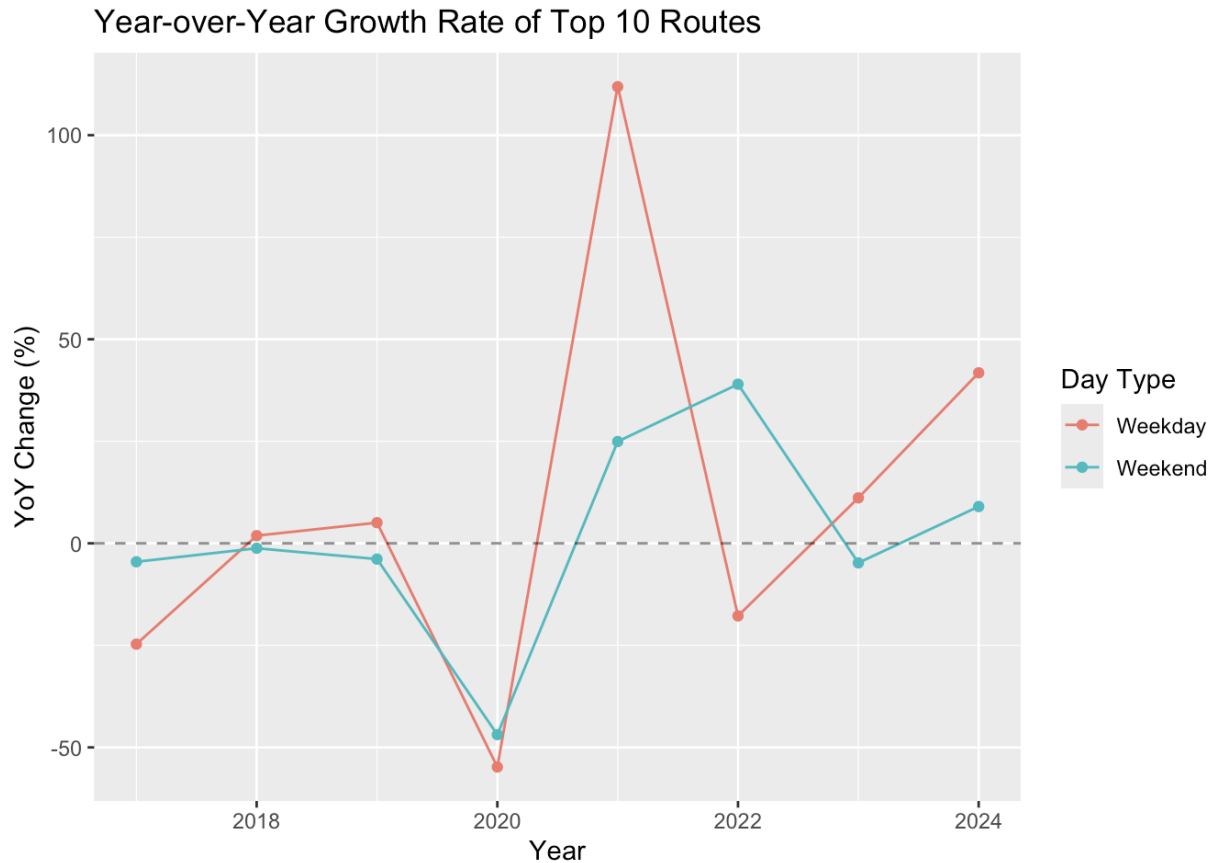
because weekend ridership is more sensitive to changes in leisure travel and tourism, whereas weekday demand remains anchored to recurring commuting patterns. Overall, the heatmap highlights both the stability of key routes and the shifts caused by COVID-19.



Analysis 5: Year-over-Year Growth Rate

We analyze the year-over-year growth rate of average loads for the top 10 routes from 2016 to 2024. Before 2020, both weekday and weekend routes show relatively small positive or negative changes, which shows that ridership on the busiest routes remained generally stable in the years leading up to the pandemic. In 2020, both lines showed a dramatic decline of nearly 50%, reflecting the severe disruption caused by COVID-19. Since 2021, both weekday and weekend ridership show strong rebounds, with weekdays in particular experiencing a sharp increase due to recovery from the unusually low 2020 baseline. This suggests that essential travel and work-related trips resumed earlier and more rapidly. In contrast, weekend growth is steadier but less extreme, indicating that leisure and discretionary travel returned more gradually. By 2023 and 2024, the growth rates for both day types stabilize around modest

positive values, implying that ridership has entered a slower, more incremental recovery phase.



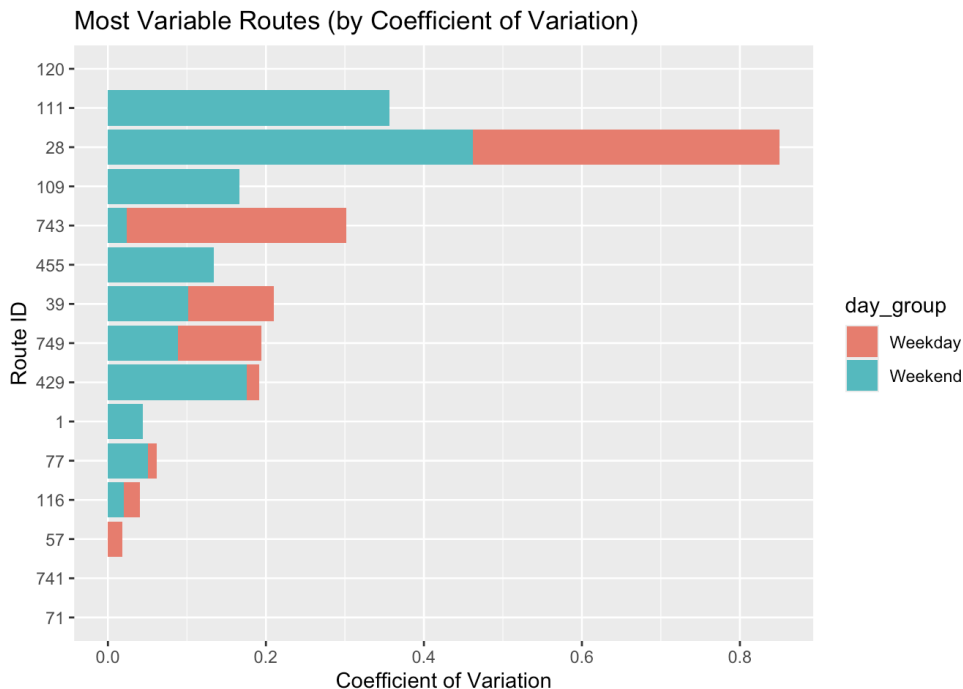
Analysis 6: Recovery Rate Comparison

We compare ridership changes across three key years, which are 2019 (pre-COVID baseline), 2020 (pandemic crash), and 2024 (latest available year) to evaluate how strongly the Top 10 routes have recovered, separated by weekdays and weekends. Before COVID, weekday routes carried slightly higher loads (74.14) than weekend routes (64.70), which shows stronger commuting demand. In 2020, both day types experienced a sharp decline, weekday averages dropped to 33.51 and weekend averages to 34.37, which is roughly a 50% reduction from 2019. By 2024, weekday routes not only recovered but exceeded their pre-COVID level, rising to 91.97, while weekend routes climbed back to 61.94, which is close to the 2019 value. This pattern shows that work-related travel has more than rebounded, whereas leisure and discretionary trips have nearly returned. Overall, weekday demand has become even stronger than before the pandemic, while weekend ridership remains somewhat subdued.

period <chr>	day_group <chr>	year_2019 <dbl>	year_2020 <dbl>	year_2024 <dbl>	covid_drop <dbl>	recovery_rate <dbl>
Pre-COVID	Weekday	74.14	NA	NA	NA	NA
Pre-COVID	Weekend	64.70	NA	NA	NA	NA
COVID-2020	Weekday	NA	33.51	NA	NA	NA
COVID-2020	Weekend	NA	34.37	NA	NA	NA
Post-Recovery	Weekday	NA	NA	91.97	NA	NA
Post-Recovery	Weekend	NA	NA	61.94	NA	NA

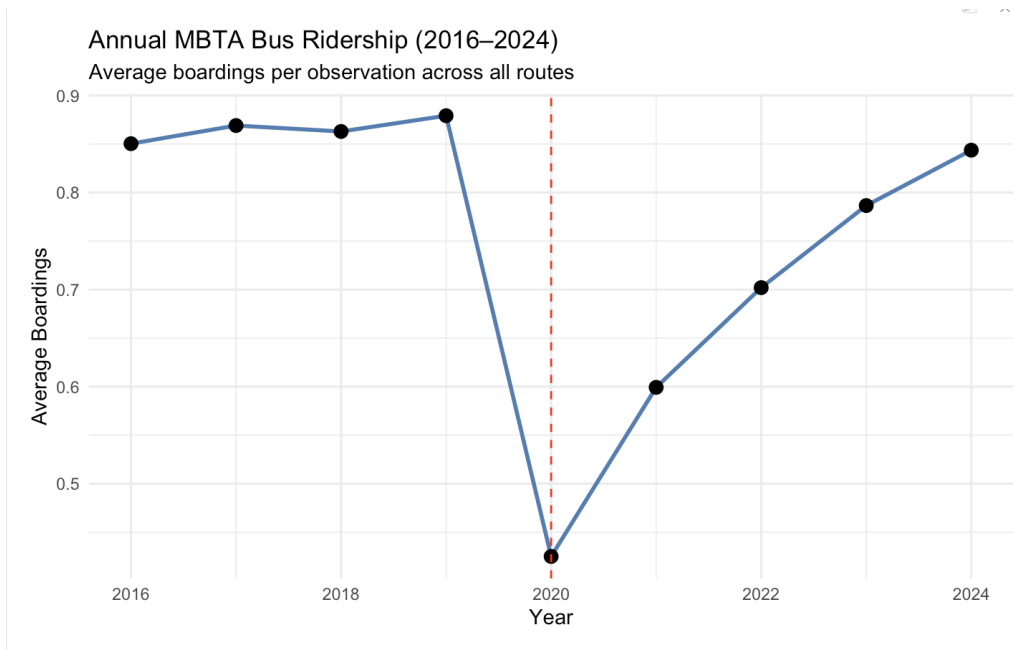
Analysis 7: Load Variability (Stability Analysis)

We rank the Top 10 routes by their coefficient of variation, which measures how unstable a route's load is over time. A higher CV means greater fluctuations and therefore a route that is more sensitive to external shocks or year-to-year changes. Routes such as 28, 111, and 120 show the highest variability, especially on weekdays, suggesting that they are more exposed to shifts in commuting patterns, neighborhood-specific disruptions, or pandemic-related demand drops. In contrast, routes with smaller CV values—such as 1, 71, or 741—maintain more consistent ridership and appear more resilient to change. Weekend values generally show less variation than weekday ones, which shows that weekend travel patterns are more stable over time. Overall, the analysis reveals that while some high-demand corridors fluctuate widely, others remain steady anchors of the MBTA bus network.



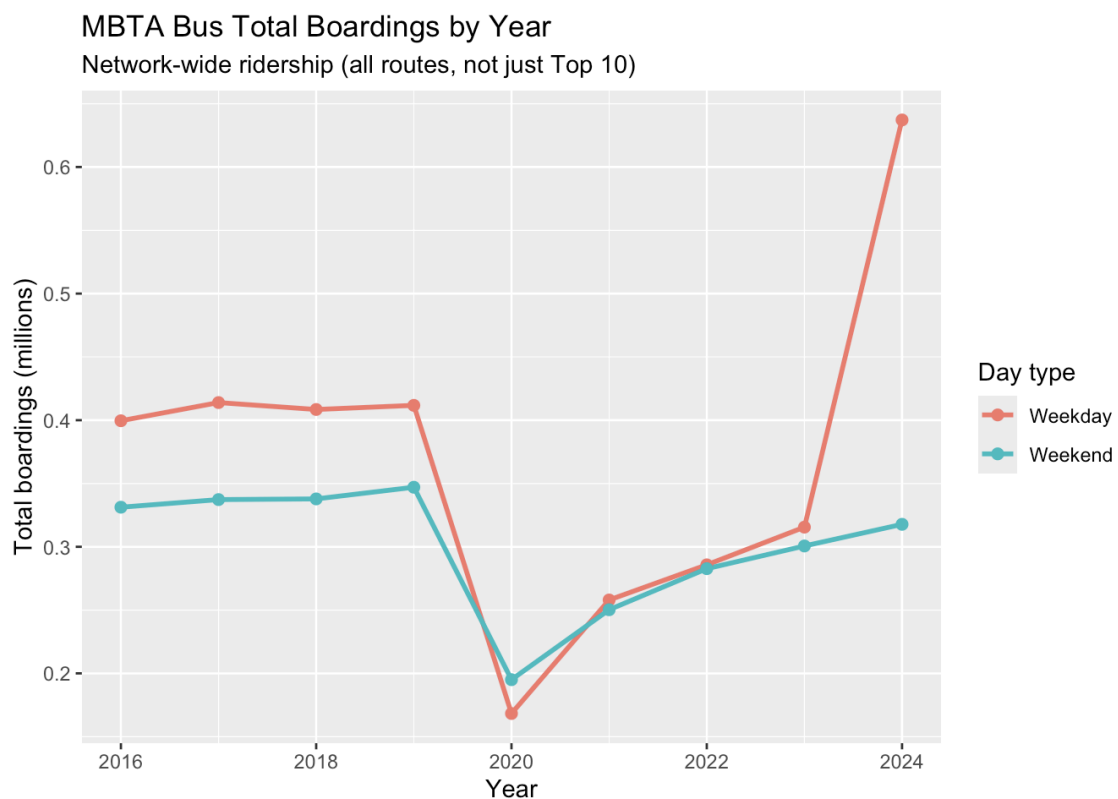
Analysis 8: Annual Trend of Ridership

We analyze annual MBTA bus ridership using Fall semester data from 2016–2024, summarizing systemwide trends through the average number of boardings per observation across all routes. Ridership remained highly stable from 2016 to 2019, with average boardings consistently around 0.85–0.88 per stop-level observation. This period reflects predictable pre-pandemic commuting patterns driven by in-person work, regular university schedules, and a stable urban travel structure. In 2020, ridership experienced a sharp decline of nearly 50%, dropping to an average of about 0.45 boardings. This collapse aligns with the onset of COVID-19, during which stay-at-home advisories, widespread remote work, and reduced campus activity significantly reduced travel demand. A gradual recovery began in 2021, with average boardings increasing each year. By 2024, ridership had returned to roughly 0.84—approximately 90–95% of its 2019 level. Although the system has recovered substantially, this partial rebound indicates that MBTA bus usage has not returned fully to pre-COVID demand. Factors such as hybrid work arrangements, more flexible academic formats, and increased use of alternative transportation modes may have reshaped long-term travel behavior. Overall, the annual trend suggests that COVID-19 produced a lasting structural shift rather than a temporary shock. MBTA bus demand appears to be stabilizing slightly below its pre-pandemic baseline, a pattern consistent with broader post-COVID urban mobility trends.



Analysis 9: Network-Wide Total Ridership Trends (2016–2024)

This chart presents the total number of passengers boarding the entire bus system of the Boston Transit Authority from 2016 to 2024, with distinctions made between weekdays and weekends. From 2016 to 2019, both sets of data were relatively stable, with only minor fluctuations, reflecting the stable passenger demand of a mature public transportation system. However, in 2020, the total number of passengers boarding on both weekdays and weekends dropped significantly, highlighting the unprecedented impact brought about by the COVID-19. Especially on weekdays, the decline was particularly severe, indicating that the commuting mode that used to be the main mode of transportation for the Boston Transit Authority passengers was severely impacted by the shift to remote working. After 2020, ridership begins a slow but steady recovery. The increase in the ridership traveling on weekdays is much greater than that on weekends. By 2024, the ridership on weekdays has a significant growth, even surpassing the pre-pandemic level, which might be caused by the return to onsite work and adjustments in public transportation service models. The ridership on weekends has also recovered, but the recovery speed is significantly slower, which shows that the regular pattern of leisure activities is still changing, the hybrid working model is also being adjusted, and the service frequency on weekends may also have decreased.

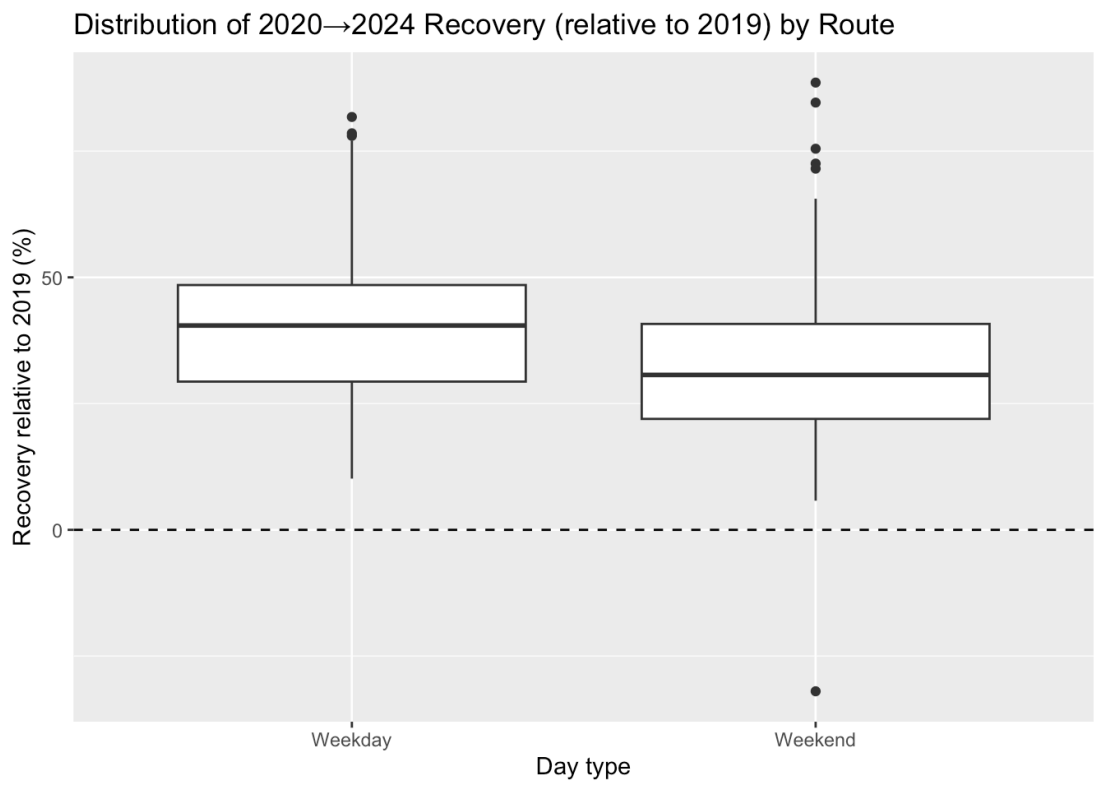
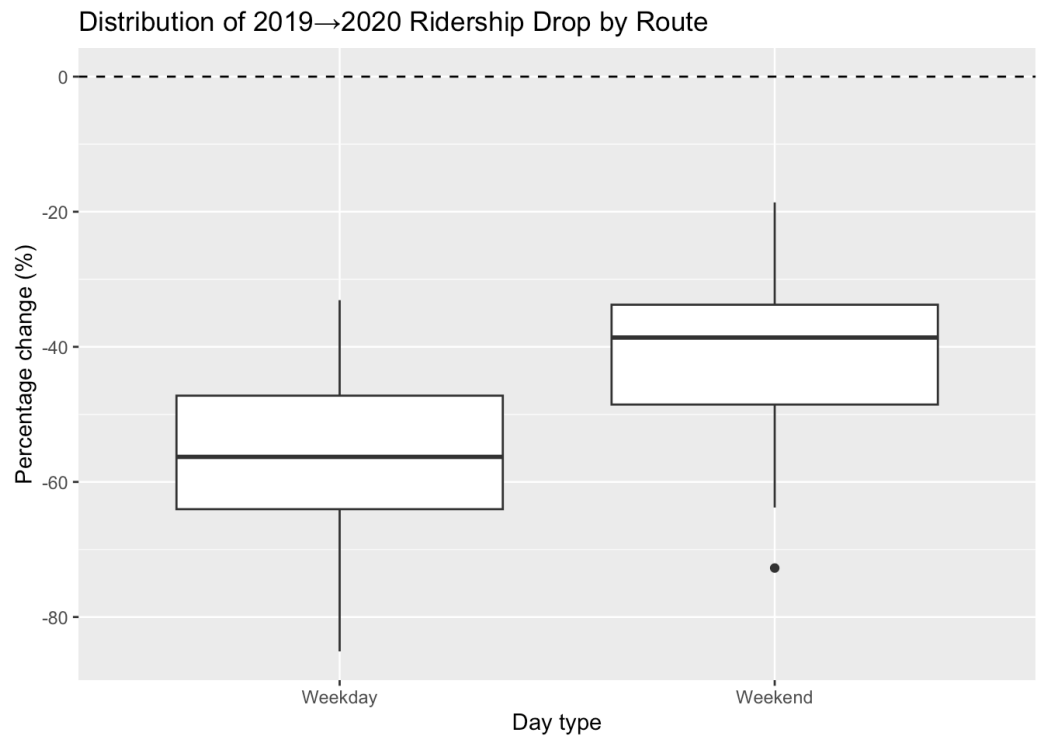


**Analysis 10: The "decline & recovery difference" between weekday and Weekend
(distributed by route, not the average value)**

Analysis 11: Route-Level Ridership Decline and Recovery

The first boxplot compares route-level ridership drops from 2019 to 2020, capturing how severely individual bus lines were affected at the start of the COVID-19 pandemic. Both weekday and weekend services experienced sharp declines, but weekday routes show a noticeably larger median drop, which is around –60% compared with weekends which is around –40%. The wider spread and deeper lower whiskers for weekday service indicate that many commuter-oriented routes collapsed almost entirely when school closures, and reduced commuting eliminated most peak-hour travel.

The second boxplot visualizes recovery from 2020 to 2024 relative to the 2019 baseline. Weekday ridership shows slightly stronger recovery—median recovery reaches around 40–45% of pre-pandemic levels, whereas weekend routes recover slightly less (around 30–35%). This pattern aligns with the gradual return to in-person work, which boosts weekday usage more directly. Weekend recovery remains weaker and more inconsistent, reflecting persistent changes in leisure travel, tourism, and discretionary mobility. Overall, the Weekday routes suffered a larger initial shock, making them more sensitive to pandemic disruptions; Weekday recovery is stronger but still far from full restoration, suggesting structural shifts in travel behavior continue to limit the ability of the system to return to 2019 levels.



Reference:

[1] Markos, M. (2021, March 8). Life in Lockdown: A Timeline of the COVID Shutdown in Massachusetts. NBC Boston.

<https://www.nbcboston.com/life-in-lockdown/life-in-lockdown-a-timeline-of-the-covid-shutdown-in-massachusetts/2320541/>

[2] Moovit. (2025). Line Route 39 - MBTA - Bus Schedules | Moovit. Moovit.

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